Estimation Methods

Xiao-Hua Zhou

Peking University

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Outline

- Three methods of constructing estimators:
  - Method of Moments (MOM)
  - Maximum likelihood (MLE)
  - Bayesian estimators (more later).
- Methods for evaluating estimators:
  - Bias and Variance
  - Mean squared error (MSE)
  - Minimax Theory
  - Large sample theory.

Parametric Point Estimation

- $X_1, \ldots, X_n \sim f(x \mid \theta)$: a random sample from the population.
- Goal: estimate θ, based on this sample.
- Definition: A point estimator is any function $W(X_1, \ldots, X_n)$ of a sample, that is, any statistic is a point estimator.
- Difference between an estimate and an estimator: an estimator is a function of the random variables, $X_1, \ldots, X_n$, and an estimate is a function of the realized values $x_1, \ldots, x_n$.

Method of Moments

- Let $\theta = (\theta_1, \ldots, \theta_k)$. Define

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i\,, \quad \mu_1(\theta) = E(X_i)$$
$$\vdots \qquad\qquad\qquad\qquad \vdots$$
$$m_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k\,, \quad \mu_k(\theta) = E(X_i^k)$$

- Let $\widehat{\theta} = (\widehat{\theta}_1, \ldots, \widehat{\theta}_k)$ solve

$$m_j = \mu_j(\widehat{\theta}), j = 1, \ldots, k.$$

- In other words, we equate the first k sample moments with the first k theoretical moments. This defines k equations with k unknowns.

Example 1

- $X_i \sim N(\beta, \sigma^2)$ with $\theta = (\beta, \sigma^2)$, $i = 1, \ldots, n$.
- MOMs:

$$\widehat{\beta} = \bar{X}, \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Example 2

- $X_i \sim Binomial(k, p)$ with $\theta = (k, p)$, $i = 1, \ldots, n$.
- MOMs:

$$\widehat{p} = \frac{\bar{X}}{\widehat{k}}, \widehat{k} = \frac{\bar{X}^2}{\bar{X} - \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

- These estimators are not the best ones as $\widehat{k}$ could be negative.

Maximum Likelihood

- The most popular method for estimating parameters is maximum likelihood.
- The reason is that, under certain conditions, the maximum likelihood estimator is optimal. This result was established by Sir Ronald Fisher and Lucian LeCam.
- The maximum likelihood estimator (mle) θ is defined as the maximizer of the likelihood,

$$L(\theta \mid x) = p(x_1, \ldots, x_n; \theta) = (i.i.d.) \prod_{i=1}^{n} p(x_i; \theta).$$

- Equivalently, the maximizer of the log-likelihood,

$$l(\theta \mid x) = logL(\theta \mid x).$$

Maximum Likelihood

- Two problems associated with finding the maximum of a function.
  - How to find a global maximum and verify that it is indeed a global maximum.
  - Numerical sensitivity: how sensitive is the estimate to small changes in the data?

Maximum Likelihood

- If the likelihood function is differentiable in θ, possible candidates are values of θ that solve

$$\frac{\partial l(\theta \mid x)}{\partial \theta_j} = 0, j = 1, \ldots, k.$$

- Note that the solutions to the above equations are only possible candidates for the MLE since the first derivative being 0 is only a necessary condition for a maximum, not a sufficient condition.

Maximum Likelihood

- Furthermore, the zeros of the first derivative locate only the extreme points in the interior of the domain of a function.
- If the extrema occur on the boundary the first derivative may not be zero. Hence, the boundary must be checked separately for extrema.

Maximum Likelihood

- Points at which the first derivatives are 0 may be local or global minima, local or global maxima or inflection points.
- Our job is to find a global maximum.

Example 2

- $X_1, \ldots, X_n \sim N(\theta, 1)$, show the MLE is $\widehat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$.
- Method 1: setting $\frac{\partial L(\theta|x)}{\partial \theta} = 0$ yields $\widehat{\theta} = \bar{x}$, which is a candidate for the MLE.
- To verify $\bar{x}$ is indeed a global maximum of the likelihood function, we can use the following argument. First, note that $\bar{x}$ is the only solution to the equation where the first derivative of the log-likelihood function is zero.

Example 2

- Second, we verify that

$$\frac{d^2}{d\theta^2} L(\theta \mid x) \mid \theta = \bar{x} < 0.$$

  Hence, $\bar{x}$ is the only extreme point in the interior and it is a maximum.

- To verify that $\bar{x}$ is a global maximum, we must check the boundaries, $+\infty$ and $-\infty$. It is easy to show that the likelihood is 0 at $\theta = \infty, -\infty$.

- Hence, $\bar{x}$ is a maximum.

Another for finding MLE

- Another way to find the MLE is to directly maximize the likelihood function.
- One general technique is to find a global upper bound on the likelihood function and then establish that there is a unique point for which the upper bound is attained.

Example 3

- Let $X_1, \ldots, X_n \sim N(\theta, 1)$.
- Recall that

$$\sum_{i=1}^{n}(x_i - a)^2 \geq \sum_{i=1}^{n}(x_i - \bar{x})^2.$$

  with equality if and only if $a = \bar{x}$.
- This implies that for any θ,

$$e^{-(1/2)\sum_i (x_i-\theta)^2} \leq e^{-(1/2)\sum_i (x_i-\bar{x})^2}$$

  Hence, $\bar{X}$ is the MLE.

Example 4

- Let $X_1, \ldots, X_n \sim$ Uniform$(0, \theta)$. Then,

$$L(\theta) = \frac{1}{\theta} I(\theta > X_{(n)}).$$

- So, $\widehat{\theta} = X_{(n)}$.

Restricted MLE

- Let $X_1, \ldots, X_n \sim N(\theta\ 1)$, where $\theta > 0$.
- Without any restriction, the MLE is $\overline{X}$.
- When $\bar{x}$ is negative, we can show that $L(\theta \mid x)$ is decreasing in θ for $\theta \geq 0$ and is maximized at $\widehat{\theta} = 0$.
- So, $\widehat{\theta} = \bar{X}$ if $\overline{X} \geq 0$ and $\widehat{\theta} = 0$ if $\overline{X} < 0$.

MLE, continued

- If $L(\theta \mid x)$ cannot be maximized analytically, it may be possible to use a computer and maximize $L(\theta \mid x)$ analytically.
- In this case, it is still a question of whether a local or global maximum has been found.

Example 5

- Let $X_1, \ldots, X_n \sim Binomial(k, p)$, where k is unknown and p is known.
- The likelihood function is

$$L(k \mid z, p) = \prod_{i=1}^{n} \binom{k}{x_i} p^{x_i} (1-p)^{(k-x_i)}.$$

- Maximizing $L(k \mid x, p)$ by differentiation is difficult because of the factorials and because k must be an integer. We try a different approach.
- Note that $L(k \mid x, p) = 0$ if $k < max_i x_i = x_{(n)}$.
- Thus, the MLE is an integer $k \geq x_{(n)}$ and satisfies that

$$L(k \mid x, p)/L(k-1 \mid x, p) \geq 1, L(k+1 \mid x, p)/L(k \mid x, p) < 1.$$

Example 5, continued

- Notice that the ratio of likelihoods is:

$$\frac{L(k \mid x, p)}{L(k-1 \mid x, p)} = \frac{(k(1-p))^n}{\prod_{i=1}^{n}(k - x_i)}.$$

- Thus the condition for a maximum is

  $(k(1-p))^n \geq \prod_{i=1}^{n}(k - x_i)$

  and $((k+1)(1-p))^n < \prod_{i=1}^{n}(k+1-x_i)$.

- Or,

$$(1-p)^n \geq \prod_{i=1}^{n}(1 - \frac{x_i}{k})$$

  and

$$(1-p)^n < \prod_{i=1}^{n}(1 - \frac{x_i}{k+1}).$$

Example 5, continued

- Since $k \geq x_{(n)}$,

$$\prod_{i=1}^{n}(1-\frac{x_i}{k}) < \prod_{i=1}^{n}(1-\frac{x_i}{k+1})$$

If k_0 is the solution that solves the following equation for k:

$$(1-p)^n = \prod_{i=1}^{n}(1-\frac{x_i}{k})$$

for $k \geq x_{(n)}$, then k_0 is the MLE for k, because

$$L(k_0 \mid x, p)/L(k_0-1 \mid x, p) \geq 1, L(k_0+1 \mid x, p)/L(k_0 \mid x, p) < 1.$$

Example 5, continued

- Note the following equation:

$$(1-p)^n = \prod_{i=1}^{n}(1-x_i z_i)$$

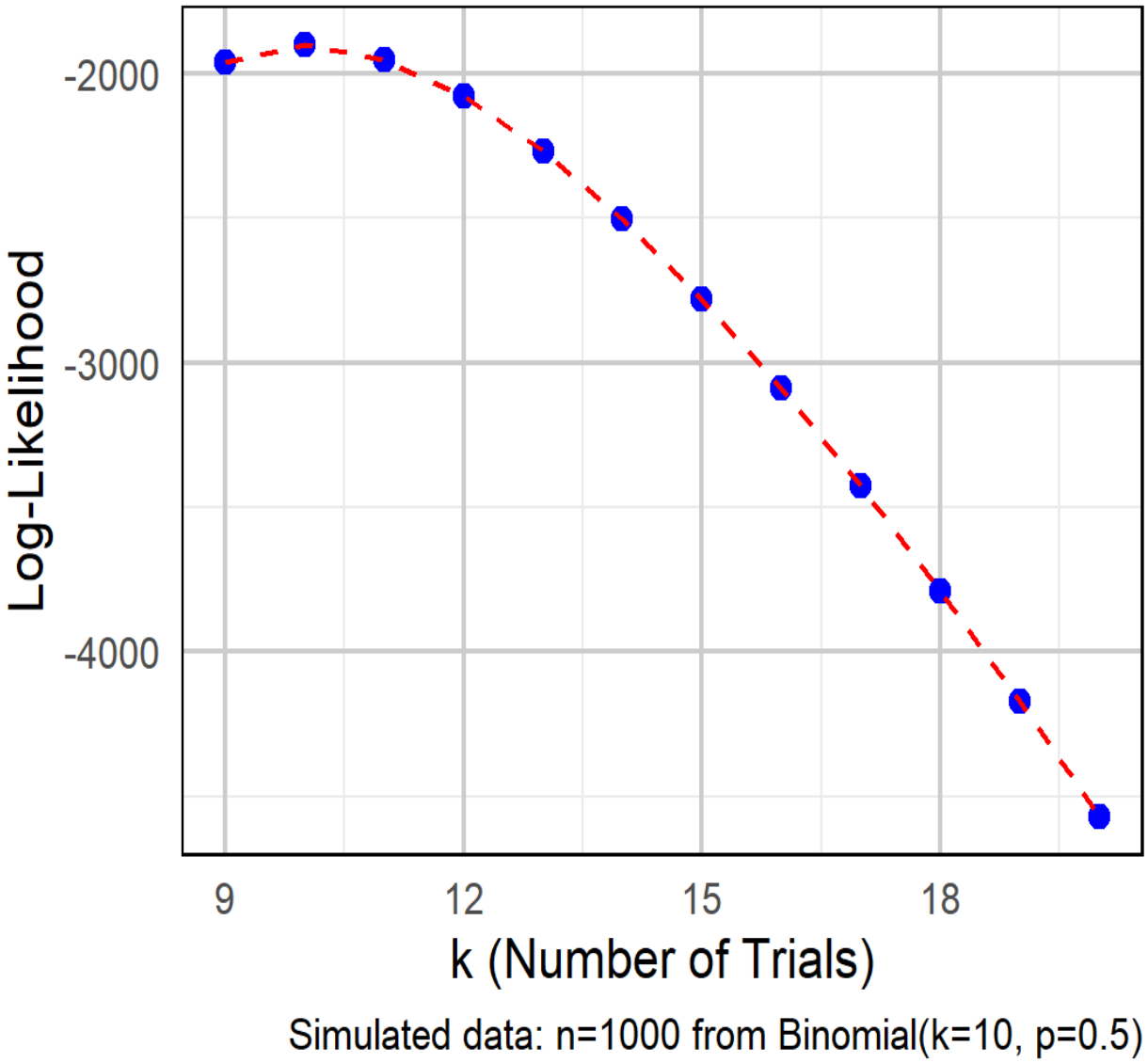
  where $z = 1/k$, for $0 \leq z \leq 1/max_i(x_i)$

- The right-hand side is a decreasing function of z for z in this range with a value of 1 at $z = 0$ and a value of 0 at $z = 1/max_i x_i$.

Example 5, continued

- Thus there is a unique z (call it $\hat{z}$) that solves the equation. The quantity $1/\hat{z}$ may not be an integer. But the integer $\hat{k}$ that satisfies the inequalities, and is the MLE, is the latest integer less than or equal to $1/\hat{z}$.
- This analysis shows that there is a unique maximum for the likelihood function and can be found by numerically solving an nth-degree polynomial equality.

Example 5, continued



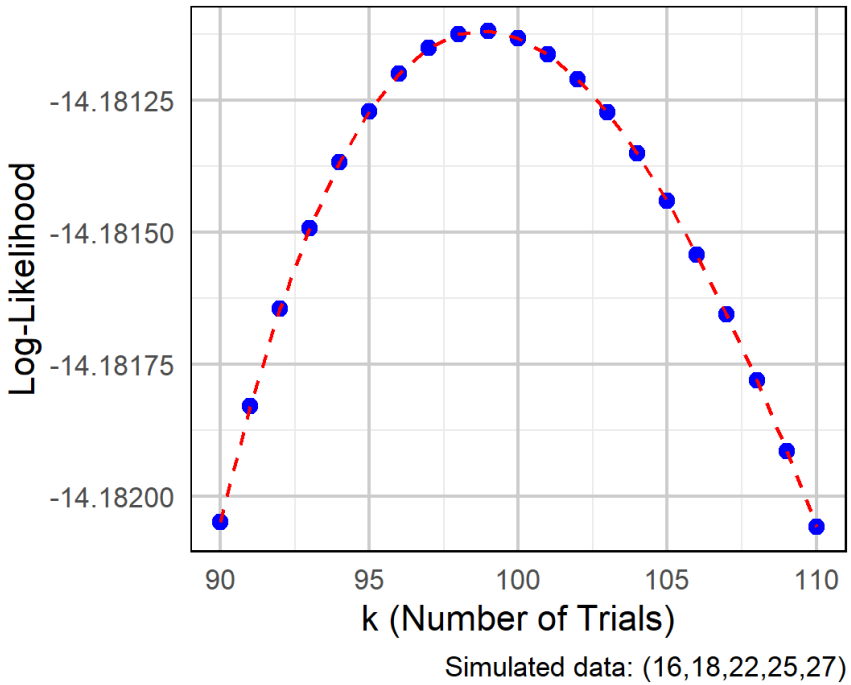
Stability of MLE

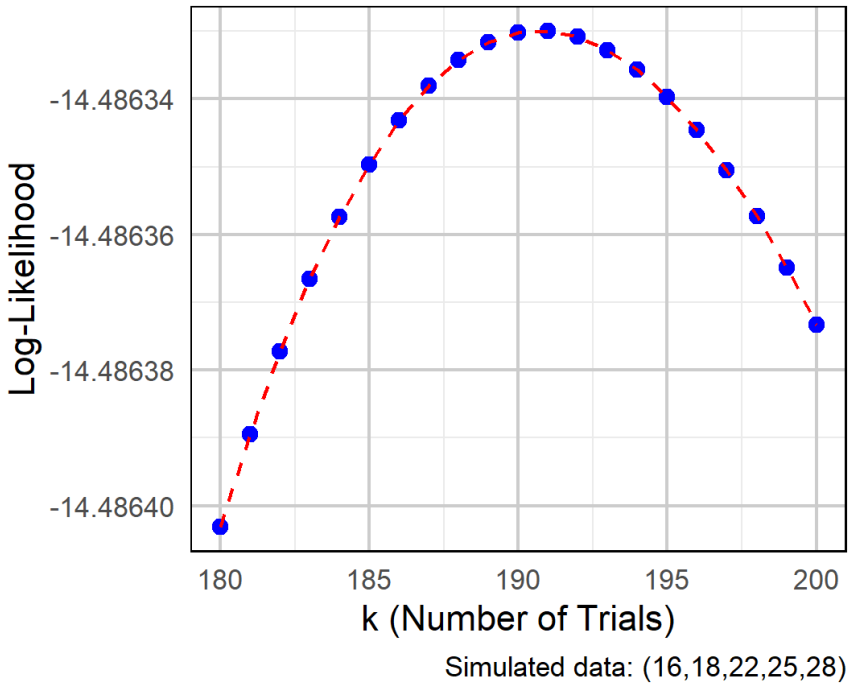
- MLE is found by a maximization process, which is susceptible to the problem associated with that process, among them that of numerical instability.
- Since the data are measured with error, we might ask how small changes in the data might affect the MLE.
- That is, we calculate $\widehat{\theta}$, based on $L(\theta \mid x)$, but we might inquire what value we would get for the MLE if we base our calculations on $L(\theta \mid x + \epsilon)$, for small ϵ. Intuitively, this new MLE should be close to $\widehat{\theta}$. But this is not always the case.

Stability of MLE

- $X_1, \ldots, X_n \sim binomial(k, p)$, where both k and p are unknown.
- Olkin et al (1981) demonstrated that the MLEs of k and p are not stable. For example, if the observed numbers of successes from an unknown number of binomial trials are (16,18,22,25,27), the MLE of k is $\widehat{k} = 99$.
- The second data set is (16,18,22,25,28), then the MLE of k is $\widehat{k} = 190$, demonstrating a large amount of variability.

Stability of MLE





Reasons

- Such occurrences happen when the likelihood function is very flat in the neighborhood of its maximum or when there is no finite maximum.
- It is often wise to spend a little time investigating the stability of the solution.

Equivariance of MLE

- $\eta = g(\theta)$.
- Let the MLE of θ be $\widehat{\theta}$.
- Then, the MLE of η is $\widehat{\eta} = g(\widehat{\theta})$.
- Proof: Suppose g is invertible. Then, $\theta = g^{-1}(\eta)$. Define $L^*(\eta) = L(\theta)$, where $\theta = g^{-1}(\eta)$
- So, for any η,

$$L^*(\widehat{\eta}) = L(\widehat{\theta}) \geq L(\theta) = L^*(\eta).$$

Hence, $\widehat{\eta} = g(\widehat{\theta})$ maximizes $L^*(\eta)$.

Proof, continued

- If g is non-invertible, for a given value η, there may be more than one value of θ that satisfies $g(\theta) = \eta$. Hence, the correspondence between the maximization over η and over θ can break down.
- We first define the induced likelihood function for $g(\theta)$: Define

$$L^*(\eta) = sup_{\theta:g(\theta)=\eta} L(\theta)$$

- Let $\widehat{\eta}$ maximize $L^*(\eta \mid x)$, which is called the MLE of $\eta = g(\theta)$.
- We need to show that $L^*(\widehat{\eta} \mid x) = L^*(g(\widehat{\theta}) \mid x)$.

Proof, continued

- For any η, by the definition of $\widehat{\eta}$,

$$L^*(\widehat{\eta}) = \sup_{\eta} \sup_{\theta : g(\theta) = \eta} L(\theta \mid x) = \sup_{\theta} L(\theta) = L(\widehat{\theta})$$

  where the second equality follows because the iterated maximization is equal to the unconditional maximization over θ, which is attained at $\widehat{\theta}$.

Proof, continued

- Furthermore,

$$L(\widehat{\theta} \mid x) = sup_{\theta:g(\theta)=g(\widehat{\theta})} L(\theta \mid x) = L^*(g(\widehat{\theta}) \mid x)$$

- Hence, we have shown that $L^*(\widehat{\eta} \mid x) = L^*(g(\widehat{\theta}) \mid x)$, which has shown that $g(\widehat{\theta})$ is the MLE of $g(\theta)$.

Example 6

- Let $X_1, \ldots, X_n \sim Binomial(n, p)$.
- the MLE of p is: $\widehat{p} = \bar{X}$.
- Let $\eta = log(p/(1-p))$. Then, the MLE of $\widehat{\eta} = log(\widehat{p}/(1-\widehat{p}))$.

Multivariate-case; Equivalence

- If the MLE of $(\theta_1, \ldots, \theta_k)$ is $(\widehat{\theta}_1, \ldots, \widehat{\theta}_k)$, and if $\tau(\theta_1, \ldots, \theta_k)$ is any function of the parameters, the MLE of $\tau(\theta_1, \ldots, \theta_k)$ is $\tau(\widehat{\theta}_1, \ldots, \widehat{\theta}_k)$

Example 7.2.11

- $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where both μ and σ are unknown.
- The likelihood:

$$L(\mu, \sigma \mid x) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(1/2)\sum_{i=1}^{n}(x_i-\mu)^2/\sigma^2}$$

- The log-likelihood function:

$$l(\mu, \sigma \mid x) = -\frac{n}{2} log(2\pi) - \frac{n}{2} log(\sigma^2) - \frac{1}{2}\sum_{i=1}^{n}(x_i - \mu)^2/\sigma^2.$$

- The partial derivatives, with respect to μ and σ^2:

$$\frac{\partial}{\partial\mu} l(\mu, \sigma \mid x) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

$$\frac{\partial}{\partial\sigma^2} l(\mu, \sigma \mid x) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Example 7.2.11

- Verifying that our solution is the global MLE is reduced to a one-dimensional problem, verifying that $(\sigma^2)^{-n/2} exp(-\frac{1}{2}\sum_{i=1}^{n}(x_i - \bar{x})^2/\sigma^2)$ achieves the global maximum at $\widehat{\sigma^2} = n^{-1}\sum_i (x_i - \bar{x})^2$.
- This is easy to show this result using the univariate calculus.
- Hence, we have shown that $(\bar{X}, n^{-1}\sum_{i=1}^{n}(X_i - \bar{X})^2)$ is the MLE.

The Second Method of Proof

- With the method of two-variate calculus, to verify a function $H(\theta_1, \theta_2)$ has a local maximum at $(\widehat{\theta}_1, \widehat{\theta}_2)$, we need to show the following three conditionals hold. (1) The first-order partial derivatives are 0, $\frac{\partial}{\partial \theta_1} H(\theta_1, \theta_2) \mid_{\theta_1 = \widehat{\theta}_1, \theta_2 = \widehat{\theta}_2} = 0$, $\frac{\partial}{\partial \theta_2} H(\theta_1, \theta_2) \mid_{\theta_1 = \widehat{\theta}_1, \theta_2 = \widehat{\theta}_2} = 0$; (2) at least one second-order partial derivative is negative, $\frac{\partial^2}{\partial \theta_1^2} H(\theta_1, \theta_2) \mid_{\theta_1 = \widehat{\theta}_1, \theta_2 = \widehat{\theta}_2} < 0$ or $\frac{\partial^2}{\partial \theta_2^2} H(\theta_1, \theta_2) \mid_{\theta_1 = \widehat{\theta}_1, \theta_2 = \widehat{\theta}_2} < 0$

The Second Method of Proof

- (3) the Jacobian of the second-order partial derivatives is positive,

$$det\left(\begin{array}{cc} \frac{\partial^2}{\partial\theta_1^2}H(\theta_1,\theta_2) & \frac{\partial^2}{\partial\theta_1\theta_2}H(\theta_1,\theta_2) \\ \frac{\partial^2}{\partial\theta_1\theta_2}H(\theta_1,\theta_2) & \frac{\partial^2}{\partial\theta_2^2}H(\theta_1,\theta_2) \end{array}\right) \mid_{\theta_1=\widehat{\theta}_1,\theta_2=\widehat{\theta}_2}$$

$$= \frac{\partial^2}{\partial\theta_1^2}H(\theta_1,\theta_2)\frac{\partial^2}{\partial\theta_2^2}H(\theta_1,\theta_2) - (\frac{\partial^2}{\partial\theta_1\theta_2}H(\theta_1,\theta_2))^2 \mid_{\theta_1=\widehat{\theta}_1,\theta_2=\widehat{\theta}_2} > 0$$

The Second Method of Proof

- (4) So far we have shown that the solution $\widehat{\theta}_1$ and $\widehat{\theta}_2$ is an interior ML.
- We still need to check that it is unique and that there is no maximum at infinity.

The Second Method of Proof

- For the normal example, we obtain that

$$\frac{\partial^2}{\partial \mu^2} l(\mu, \sigma^2) = -\frac{n}{\sigma^2},$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} l(\mu, \sigma^2) = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^{n} (x_i - \mu)^2,$$

$$\frac{\partial^2}{\partial \mu \sigma^2} l(\mu, \sigma^2) = -\frac{1}{\sigma^4} \sum_{i=1}^{n} (x_i - \mu).$$

- The Jacobian is $\frac{1}{\widehat{\sigma}^6} \frac{n^2}{2} > 0$

Cramer-Rao Inequality about the Variance

- Theorem 7.3.9: Let $X_1, \ldots, X_n$ be i.i.d. with density function $f(x \mid \theta)$. Let $W = W(X_1, \ldots, X_n)$ be any estimate of θ, satisfying

$$\frac{\partial}{\partial \theta} E_\theta W(X) = \int \frac{\partial}{\partial \theta} [W(x) f(x \mid \theta) dx]$$

and $Var_\theta W(X) < \infty$.

- Then, under smoothness assumptions on $f(x \mid \theta)$,

$$Var(W(X)) \geq \frac{(\frac{\partial}{\partial \theta} E_\theta W(X))^2}{E_\theta((\frac{\partial}{\partial \theta} log f(X \mid \theta))^2)} = \frac{(\frac{\partial}{\partial \theta} E_\theta W(X))^2}{I(\theta)},$$

where $I(\theta)$ is called Fisher information matrix.

Proof

- The proof is a cleaver application of Cauchy-Schwarz Inequality, which says that for any two random variables X and Y,

$$[Cov(X, Y)]^2 \leq (VarX)(VarY).$$

- Rearranging the above inequality, we obtain

$$Var(X) \geq \frac{[Cov(X, Y)]^2}{Var(Y)}.$$

- The cleverness in the proof is that we choose X to be $W(X)$ and Y to be $\frac{\partial}{\partial\theta} logf(X \mid \theta)$ in the Cauchy-Schwarz Inequality.

Proof

- Here are some facts about these two variables.

$$(1)\frac{\partial}{\partial\theta}E_\theta W(X) = \int W(x)[\frac{\partial}{\partial\theta}f(x \mid \theta)]dx =$$

$$E_\theta[W(X)\frac{\partial}{\partial\theta}logf(x \mid \theta)]$$

  that suggests a covariance between $W(X)$ and $\frac{\partial}{\partial\theta}logf(X \mid \theta)$,which requires us to subtract the product of the expected values.

- If we apply (1) with $W(X) = 1$, we obtain $E_\theta[\frac{\partial}{\partial\theta}logf(x \mid \theta)] = \frac{\partial}{\partial\theta}E_\theta 1 = 0$
- Hence, using the result in (1) again, $Cov_\theta(W(X), \frac{\partial}{\partial\theta}logf(x \mid \theta)) = \frac{\partial}{\partial\theta}E_\theta W(X)$

Proof

- Since $E_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta)) = 0$, we have

$$Var_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta)) = E_\theta((\frac{\partial}{\partial\theta} logf(X \mid \theta))^2).$$

- Using the Cauchy-Schwarz Inequality, we obtain

$$Var_\theta(W(X)) \geq \frac{(\frac{\partial}{\partial\theta} E_\theta W(X))^2}{E_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta))^2}$$

  This completes the proof.

- The quantity $E_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta))^2$ is called the information matrix or Fisher information matrix.

Corollary one

- If $X_1, \ldots, X_n$ are i.i.d., and the assumptions of Theorem 7.3.9 are satisfied, then

$$Var_\theta(W(X)) \geq \frac{(\frac{\partial}{\partial\theta} E_\theta W(X))^2}{nE_\theta(\frac{\partial}{\partial\theta} logf(X_1 \mid \theta))^2}$$

- Proof: It is easy show this result by noting

$$E_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta))^2 = \sum_{i=1}^{n} E_\theta(\frac{\partial}{\partial\theta} logf(X_i \mid \theta))^2$$

- If $W(X)$ is an unbiased estimator of θ, we have

$$Var_\theta(W(X)) \geq \frac{1}{E_\theta(\frac{\partial}{\partial\theta} logf(X \mid \theta))^2}$$

Corollary two

- If $f(x \mid \theta)$ satisfies

$$\frac{\partial}{\partial\theta}E_\theta(\frac{\partial}{\partial\theta}logf(X \mid \theta)) = \int \frac{\partial}{\partial\theta}[\frac{\partial}{\partial\theta}logf(x \mid \theta)f(x \mid \theta)]dx,$$

then

$$E_\theta(\frac{\partial}{\partial\theta}logf(X \mid \theta))^2 = -E_\theta(\frac{\partial^2}{\partial\theta^2}logf(X \mid \theta))$$

Proof: Note that

$$\frac{\partial^2}{\partial\theta^2}logf(x \mid \theta) = \frac{\frac{\partial^2}{\partial\theta^2}f(x \mid \theta)}{f(x \mid \theta)} - (\frac{\frac{\partial}{\partial\theta}f(x \mid \theta)}{f(x \mid \theta)})^2$$

Taking the expectation with respect to $f(. \mid \theta)$, we obtain the desired result.

Key Assumption

- A key assumption in the Cramer-Rao Inequality is the ability to differentiate under the integral sign, which may be restrictive.
- Example: $X_1, \ldots, X_n \sim \mathit{uniform}(0, \theta)$, the Cramer-Rao assumption does not hold.
- Leibnitz's rule: if $f(x, \theta), a(\theta), b(\theta)$ are differentiable with respect to θ, then

$$\frac{\partial}{\partial \theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx = f(b(\theta), \theta) \frac{\partial}{\partial \theta} b(\theta) - f(a(\theta), \theta) \frac{\partial}{\partial \theta} a(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx$$

Key Assumption

- Using Leibnitz's rule, we obtain

$$\frac{\partial}{\partial\theta}\int_0^\theta h(x)f(x \mid \theta)dx = \frac{\partial}{\partial\theta}\int_0^\theta h(x)\frac{1}{\theta}dx$$

$$h(\theta)/\theta + \int_0^\theta h(x)\frac{\partial}{\partial\theta}(\frac{1}{\theta})dx \neq \int_0^\theta h(x)\frac{\partial}{\partial\theta}f(x \mid \theta)dx.$$

  unless $h(\theta)/\theta = 0$ for all θ.

Multivariate Version of Cramer-Rao Inequality

Let $X_1, \ldots, X_n$ be i.i.d. with density function $f(x \mid \theta)$, where θ is a m-dimensional parameters in Ω. Let $W = W(X_1, \ldots, X_n)$ be any estimate of θ, satisfying

$$\frac{\partial}{\partial \theta} E_\theta W(X) = \int \frac{\partial}{\partial \theta} [W(x) f(x \mid \theta) dx]$$

and $Cov_\theta W(X) < \infty$.Then, under smoothness assumptions on $f(x \mid \theta)$,

$$Cov(W(X)) \geq [\frac{\partial E_\theta W(X)}{\partial \theta}]^T [I(\theta)]^{-1} [\frac{\partial E_\theta W(X)}{\partial \theta}],$$

for any $\theta \in \Omega$, and $I(\theta)$ is the Fisher information matrix.

Multivariate Version for Unbiased Estimators

- If θ is an unbiased estimator of a vector of multivariate parameters, θ, then we have the following result:

$$Cov(\widehat{\theta}) \geq \frac{1}{n} \frac{1}{Cov(\frac{1}{\partial\theta} logf(X - [i], \theta), \frac{1}{\partial\theta} logf(X - [i], \theta))},$$

where the inequality holds in the sense that the left side of the above equation minus the right is positive semi-definite.

Best Unbiased Estimators

- A shortcoming in the Cramer-Rao inequality is that there is no guarantee that the bound is sharp.
- An Example: let $X_1, \ldots, X_n$ be iid $f(. \mid \sigma^2) = N(\mu, \sigma^2)$, where μ is known. Since $\frac{\partial^2 logf(x|\sigma^2)}{\partial(\sigma^2)^2} = \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}$, $E(\frac{\partial^2 logf(x|\sigma^2)}{\partial(\sigma^2)^2}) = \frac{1}{2\sigma^4}$. Hence, for any unbiased estimator, W, of σ^2 must satisfy that $Var(W \mid \mu, \sigma^2) \geq \frac{2\sigma^4}{n}$.
- For a sample variance estimator S, which is an unbiased estimator for σ^2, we can show that $Var(S^2 \mid \mu, \sigma^2) = \frac{2\sigma^4}{n-1}$, so, S^2 does not attain the Cramer-Rao lower bound.

When is Cramer-Rao Bound Attainable

Corollary 7.3.15: Let $X_1, \ldots, X_n$ be iid $f(x \mid \theta)$, where $f(x \mid \theta)$, where $f(x \mid \theta)$ satisfies the conditions of the Cramer-Rao theorem. Let $L(\theta \mid x) = \prod_{i=1}^{n} f(x_i \mid \theta)$ be the likelihood function. If $W(X) = W(X_1, \ldots, X_n)$ is a unbiased estimator of $g(\theta)$, $W(X)$ attains the Cramer-Rao Lower Bound if and only if

$$a(\theta)[W(X) - g(\theta)] = \frac{\partial}{\partial \theta} log L(\theta \mid x).$$

for some function $a(\theta)$.

Lemma

We first show a property about correlation between X and Y:

Lemma: Let $\sigma_X = \sqrt{Var(X)}$, $\sigma_Y = \sqrt{Var(Y)}$, $\rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$, correlation between X and Y, $-1 \leq \rho_Y \leq 1$, and $\mid \rho_{XY} \mid = 1$ if and only if there exist numbers a and b such that $P(Y = aX + b) = 1$. If $\rho_{XY} = 1$, then $a > 0$ and If $\rho_{XY} = -1$, then $a < 0$.

Proof of Lemma

Proof: Consider the function,

$h(t) = E((X - \mu_X)t + (Y - \mu_Y))^2$, which is

$t^2\sigma_X^2 + 2tCov(X, Y) + \sigma_Y^2$. Since this quadratic function of t is greater than 0 or equal to 0 for all t, it can have at most one real root and must have a nonpositive discriminant for the quadratic equation $h(t)$ because it has to be a convex downward function.

Proof of Lemma, continued

- Since $h(t) = Var(X)t^2 + Var(Y) + 2tCov(X, Y)$, its discriminant is $(2Cov(X, Y))^2 - 4\sigma_X^2\sigma_Y^2$
- Hence, $(2Cov(X, Y))^2 - 4\sigma_X^2\sigma_Y^2 \leq 0$ which is equivalent to $-\sigma_X\sigma_Y \leq Cov(X, Y) \leq \sigma_X\sigma_Y$.
- Note that $\mid \rho_{XY} \mid = 1$ means that the discriminant is zero, which means that $h(t)$ has a single root. Since $[(X - \mu_X)t + (Y - \mu_Y)]^2 \geq 0$ for all X and Y, $h(t_0) = 0$ if and only if $P([(X - \mu_X)t_0 + (Y - \mu_Y)]^2 = 0) = 1$
- That is, $P(Y = aX + b) = 1$ with $a = -t_0$ and $b = \mu_X t_0 + \mu_Y$, where t_0 is the root of $h(t) = 0$.

Proof of Obtaining Cramer-Rao Inequality

- The Cramer-Rao Inequality can be written as

$$[Cov_\theta(W(X), \frac{\partial}{\partial\theta} log \prod_{i=1}^{n} f(X_i \mid \theta))]^2 \leq$$

$$Var_\theta(W(X)) Var_\theta(\frac{\partial}{\partial\theta} log \prod_{i=1}^{n} f(X_i \mid \theta))$$

- Since $E(W) = g(\theta)$ and $E_\theta(\frac{\partial}{\partial\theta} log \prod_{i=1}^{n} f(X_i \mid \theta)) = 0$, using the result in the lemma above, we can have the equality if and only if $W(X) - g(\theta)$ is proportional to $\frac{\partial}{\partial\theta} log \prod_{i=1}^{n} f(X_i \mid \theta)$.

Methods of Evaluating Estimators

- Mean squared error (MSE): $E_\theta(\widehat{\theta} - \theta)^2$:

$$\int \dots \int (\widehat{\theta}(x_1, \dots, x_n) - \theta)^2 p(x_1; \theta) \dots p(x_n; \theta) dx_1 \dots dx_n.$$

- The bias is $B = E_\theta(\widehat{\theta}) - \theta$
- Variance: $V = Var_\theta(\widehat{\theta})$.

Methods of Evaluating Estimators

- Theorem: $MSE = B^2 + V$

  Proof: see the blackboard

- An estimator is unbiased if the bias is 0. In that case, the MSE = Variance. There is often a tradeoff between bias and variance. So low bias can imply high variance and vice versa.
- Note that the MSE is a function of θ.
- Minimax estimators, that use the maximum of the MSE over θ as a way to compare estimators.

Best Unbiased Estimators

- An estimator W is a UMVUE (Uniform Minimum Variance Unbiased Estimator) for $\tau(\theta)$ if (i) $E_\theta(W) = \tau(\theta)$ for all θ and (ii) if $E_\theta(W') = \tau(\theta)$ for all θ then $Var_\theta(W) \leq Var_\theta(W')$.

Best Unbiased Estimators

- The Cramer-Rao inequality gives a lower bound on the variance of any unbiased estimator. The bound is:

$$Var_\theta(W) \geq \frac{(\frac{d}{d\theta} E_\theta W)^2}{E_\theta((\frac{\partial}{\partial\theta} logf(X;\theta))^2)} = \frac{(\tau'(\theta))^2}{I_n(\theta)}.$$

Sufficiency and Unbiaseness: Rao-Backwell Theorem

- Let W be an unbiased estimator of $\tau(\theta)$ and let T be a sufficient statistic.
- Define $W' = \phi(T) = E(W \mid T)$.
- Then W' is unbiased and

$$Var_\theta(W') \leq Var_\theta(W)$$

for all θ.

Proof of Rao-Backwell Theorem

- Since $Var(W) = Var(E(W \mid T)) + E(Var(W \mid T))$, $Var(W) \geq Var(\phi(T))$.
- It remains to show that $\phi(T)$ is indeed an estimator, which is a function only of the sample and is independent of θ.
- From the definition of sufficiency and the fact that W is a function only of the sample; it follows that the distribution of $W \mid T$ is independent of θ.

Uniqueness of Best Unbiased Estimator

- Theorem: If W is a best and unbiased estimator of $\tau(\theta)$, then W is unique.
- Proof: Suppose W' is another unbiased best estimator, and consider the estimator $W* = (W + W')/2$. Note that $E(W*) = \tau(\theta)$ and
  $Var(W*) = Var(W)/4 + Var(W')/4 + (1/2)Cov(W, W')$,
  by Cauchy-Schwarz Inequality,
  $\leq Var(W)/4 + Var(W')/4 + (1/2)\sqrt{Var(W)Var(W')} =$
  $Var(W)$, because $Var(W) = Var(W')$
- But if the above inequality is strict, the best unbiasedness of W is contradicted, so we must have equality for all θ.

Uniqueness of Best Unbiased Estimator

- Since the inequality is an application of Cauchy-Schwarz, we can have equality only if $W' = a(\theta)W + b(\theta)$. Using properties of covariance, we have $Cov(W, W') = a(\theta)Var(W)$.
- However, since $Var(W) = Var(W')$ and the equality in Cauchy-Schwarz inequality, we have that $Cov(W, W') = Var(W)$. Therefore, $a(\theta) = 1$.
- Since $E(W') = \tau(\theta)$, $b(\theta) = 0$.

- We now know that, in looking for the best unbiased estimator of $\tau(\theta)$, we need to consider only estimators based on a sufficient statistic.
- How can we find the best unbiased estimator?
- To see when an unbiased estimator is best unbiased, we might ask how could we improve an unbiased estimator?
- Let W be an unbiased estimator of $\tau(\theta)$, and let U be another estimator of 0; that is, $E(U) = 0$ for all θ.
- Then,

$$\phi_a = W + aU$$

is an unbiased estimator of $\tau(\theta)$.

- Can ϕ_a be better than W?
- $$Var_\theta(\phi_a) = Var_\theta(W) + 2aCov_\theta(W, U) + a^2 Var_\theta(U).$$
- Now if for some $\theta = \theta_0$, $Cov_{\theta_0}(W, U) < 0$, we can then make $2aCov_\theta(W, U) + a^2 Var_\theta(U) < 0$ by choosing $a \in (0, -2Cov_{\theta_0}(W, U))$.
- Hence, the resulting estimators are better than W at θ_0. As a result, W cannot be the best unbiased.
- Similarly, we can show that if $Cov_{\theta_0}(W, U) > 0$, W cannot be the best unbiased.

Finding Best Unbiased Estimators

- Theorem: If W is an unbiased estimator of $\tau(\theta)$, then W is the best unbiased estimator of $\tau(\theta)$ if and only if W is uncorrelated with all unbiased estimators of 0.
- Since an unbiased estimator of 0 is just random noise, that is, there is no information in an estimator of 0. Hence, if an estimator could be improved by adding random noise to it, the estimator probably is not good.
- This tuition is behind the above theorem.

Proof

- Proof: Suppose W' is best unbiased. Let U be another unbiased estimator of 0. Let $\phi_a = W + aU$, where a is chosen so that ϕ_a is unbiased.
- $Var(\phi_a) = Var(W) + 2aCov(W, U) + a^2 Var(U)$. If for some $\theta = \theta_0$, $Cov(W, U) < 0$, then we can make $2aCov(W, U) + a^2 Var(U) < 0$ by choosing $a \in (0, -2Cov(W, U)/Var(U))$. Hence, ϕ_a will be better than W at $\theta = \theta_0$ and W cannot be a best unbiased.
- Similarly, we can show that if $Cov(W, U) > 0$ for some θ_0, W also cannot be best unbiased. Therefore, $Cov(W, U) = 0$ for all θ; the necessity is shown

Proof for sufficient part

- Assume W is uncorrelated with all unbiased estimators of 0. Let W' be any other unbiased estimator. We need to show that W is better than W'".
- Note that we can write $W' = W + (W' - W)$. Therefore, $Var(W') = Var(W) + Var(W' - W) + 2Cov(W, W' - W)$. Since $W' - W$ is an unbiased estimator of 0 and W and W' is uncorrelated, $Cov(W, W' - W) = 0$.
- Hence, $Var(W') \geq Var(W)$. We have established the sufficiency.

Finding the Unbiased Best Estimator

- Although we have an interesting characteristic of the best unbiased estimators, it is hard to use in practice as it is hard to show that this estimator is uncorrelated with all unbiased estimators of 0.
- However, this property may be useful to determine that an estimator is not the best unbiased.

Example 7.3.21

- Let X be an observation from a uniform $(\theta, \theta + 1)$ distribution. Then, $E(X) = \theta + 1/2$. So $X - 1/2$ is an unbiased estimator of θ and $Var(X) = 1/12$.
- Let $h(X)$ be an unbiased estimator of 0, then, $\int_{\theta}^{\theta+1} h(x)dx = 0$ for all θ, and then $0 = \frac{d}{d\theta} \int_{\theta}^{\theta+1} h(x)dx = h(\theta + 1) - h(\theta)$ for all θ. Hence, $h(x)$ is a periodic function with period 1.

Example 7.3.21

- Let us take $h(x) = sin(2\pi.x)$. Then,

  $sin(2\pi(x+1)) - sin(2\pi x) = 0$

  $Cov_\theta(X - 1/2, sin(2\pi X)) = \int_\theta^{\theta+1} xsin(2\pi x)dx = -\frac{cos(2\pi\theta)}{2\pi}$

- Hence $X - 1/2$ is correlated with an unbiased estimator of zero; it cannot be the best unbiased.

Completedness

- If a family of pdfs $f(x \mid \theta)$ has the property that there are no unbiased estimators of zero (other than zero itself), then our search is over, since $Cov(W, 0) = 0$.
- To search for the best unbiased estimator, what is needed is some characterization of all unbiased estimators of zero.
- Recall that the property of completeness guarantees such a situation.

Theorem

- (a) Let T be a complete and sufficient statistics for θ, and $\phi(T)$ be any estimator only based on T. Then, $\phi(T)$ is the unique best unbiased estimator of its expected value.
- (b) If T is a complete sufficient statistic for θ, and $h(X_1, \ldots, X_n)$ is any unbiased estimator of $\tau(\theta)$, then $\phi(T) = E(h(X_1, \ldots, X_n) \mid T)$ is the best unbiased estimator of $\tau(\theta)$. (Homework, Exercise 7.56)

Example 7.3.22

- Let $X_1, \ldots, X_n \sim \mathit{uniform}(0, \theta)$, and $Y = \mathit{max}(X_i)$.
- Then, $\frac{n+1}{n} Y$ is an unbiased estimator of θ.
- Note that the conditions of the Cramer-Rao Theorem are not satisfied and cannot be applied.
- Since we have shown before that Y is a complete sufficient statistic, applying the above theorem, we have shown that $\frac{n+1}{n} Y$ is the best unbiased estimator of θ,

Example 7.3.24

- Let $X_1, \ldots, X_n \sim \textit{binomial}(k, \theta)$. .
- Question: Find an estimator for the probability of exactly one success from a $\textit{binomial}(k, \theta)$, that is,

$$\tau(\theta) = P_\theta(X = 1) = k\theta(1-\theta)^{k-1}.$$

- Note that $Y = \sum_{i=1}^{n} X_i \sim \textit{binomial}(kn, \theta)$ is a complete sufficient statistic; however, no unbiased estimator based on it is obvious.
- Let us try a simple-minded estimator, $h(X_1) = 1$ if $X_1 = 1$ and $h(X_1) = 0$ otherwise. Then, $E_\theta h(X_1) = \tau(\theta)$.

Example 7.3.24

- Our theory says that the estimator, $\phi(Y) = E(h(X_1) \mid Y)$ is the best unbiased estimator of $\tau(\theta)$.
- $\phi(t) = E(h(X_1) \mid Y = t) = P(X_1 = 1 \mid Y = t) = \frac{P_\theta(X_1=1, Y=t)}{P_\theta(Y=t)}$
  $= \frac{P_\theta(X_1=1, \sum_{i=2}^n X_i = t-1)}{P(Y=t)} = \frac{P_\theta(X_1=1) P_\theta(\sum_{i=2}^n X_i = t-1)}{P_\theta(Y=t)} = k \frac{\binom{k(n-1)}{t-1}}{\binom{kn}{t}}$
- So the best unbiased estimator of $\tau(\theta)$ is:
$$\phi(Y) = k \frac{\binom{k(n-1)}{Y-1}}{\binom{kn}{Y}}$$
- We can assert unbiasedness without performing the difficult evaluation of $E_\theta(\phi(Y))$

Asymptotic Properties of Maximum Likelihood Estimators

Under certain conditions on the model, ML estimator $\widehat{\theta}$ possesses many good properties:

- Consistency: $\widehat{\theta} \xrightarrow{P} \theta_0$, where θ_0 is the true value of θ.
- Asymptotical normality: $\sqrt{n}(\widehat{\theta} - \theta)/\widehat{se} \xrightarrow{d} N(0, 1)$.
- Asymptotic efficiency: roughly speaking, among all well-behaved estimators, the MLE has the smallest variance, at least for large samples.

Regularity Conditions

- (A0). The distributions P_θ of the observations are distinct (otherwise, θ cannot be estimated consistently).
- (A1). The distributions P_θ have common support.
- (A2). The observations are $X = (X_1, \ldots, X_n)$, where the X_i are iid with probability density $f(x \mid \theta)$ with respect to ν.
- (A3). The parameter space contains an open set ω of which the true parameter value θ_0 is an interior point.

Consistency of an MLE

- Let

$$L(\theta \mid x) = \prod_{i=1}^{n} p(x_i; \theta) \text{ and } l(\theta \mid x) = \log L(\theta \mid x)$$

- Under (A0)-(A3), and suppose for almost all x, $f(x \mid \theta)$ is differentiable with respect to $\theta \in \omega$. If the log-likelihood equation ($S(\theta \mid x) = \sum_{i=1}^{n} \frac{\partial log p(x_i;\theta)}{\partial \theta}$) has a unique root, $\widehat{\theta}_n$, for each n and all x, then $\widehat{\theta}_n$ is a consistent estimator of θ.

Heuristic Proof

- ML estimator $\widehat{\theta}$ maximizes the log-likelihood, $Q(\theta; x) = (1/n) \sum_{i=1}^{n} logp(x_i; \theta)$,$\widehat{\theta} = argmax Q(\theta; x)$. By the WLLN, we know $Q(\theta; X) \rightarrow Q_0(\theta)$ in probability, where$Q_0(\theta) = E_{\theta_0}(logp(x_i; \theta))$.
- We expect that, on average, the log-likelihood will be close to the expected log-likelihood.
- Thus, the maximum likelihood estimator will be close to the maximum of the expected log-likelihood.
- Hence we need to show that the true value θ_0 can maximize the expected log-likelihood, $Q_0(\theta)$

Lemma

If θ_0 is identified and $E_{\theta_0} \mid logp(X_i;\theta) \mid < \infty$ for all $\theta \in \Omega$, $Q_0(\theta)$ is uniquely maximized at $\theta = \theta_0$.

Proof: By Jensen's inequality, we know that for any strictly convex function $g(\cdot)$, $E[g(Y)] > g(E[Y])$. If we take $g(y) = -log(y)$. So, for $\theta \neq \theta_0$,

$$E_{\theta_0}[-log(\frac{p(X_i;\theta)}{p(X_i;\theta_0)})] > -log[E_{\theta_0}\frac{p(X_i;\theta)}{p(X_i;\theta_0)}]$$

Note that

$$E_{\theta_0}\frac{p(X_i;\theta)}{p(X_i;\theta_0)}] = \int \frac{p(X_i;\theta)}{p(X_i;\theta_0)} p(x;\theta_0)dx$$

$$= \int p(x;\theta)dx = 1$$

Proof, continued

- So,

$$E_{\theta_0}[-log(\frac{p(X_i;\theta)}{p(X_i;\theta_0)})] > 0$$

- Or,

$$Q_0(\theta_0) > Q_0(\theta)$$

- The above inequality holds for all $\theta \neq \theta_0$

Consistency of the MLE, continued

- If we can show that the limit of the maximum is the maximum of the limit, which implies that $\widehat{\theta}$ should converge in probability to θ_0.
- Sufficient conditions for the maximum of the limit to be the limit of the maximum are that the convergence is uniform and the parameter space is compact.
- In theory, compactness requires that one know bounds on the true parameter value, although this constraint is often ignored in practice.
- It is possible to drop this assumption if the function $Q(\theta, X)$ cannot rise too much as θ becomes unbounded.

Definition (Uniform Convergence in Probability)

- $Q(\theta, X)$ converges uniformly in probability to $Q_0(\theta)$ if

$$sup_{\theta \in \Omega} \mid Q(\theta; X) - Q_0(\theta) \mid \rightarrow 0 \text{ in probability } P(\theta_0)$$

- In other words, for all $\epsilon > 0$,

$$P_{\theta_0}[sup_{\theta \in \Omega} \mid Q(\theta; X) - Q_0(\theta) \mid > \epsilon] \rightarrow 0$$

Definition (Uniform Convergence in Probability)

- Why isn't pointwise convergence enough?
- Uniform convergence guarantees that for almost all realizations, the paths in θ are in the sleeve.
- This ensures that the maximum is close to θ_0.
- For pointwise convergence, we know that at each θ, most of the realizations are in the sleeve, but there is no guarantee that for another value of θ the same set of realizations is in the sleeve.
- Thus, the maximum need not be near θ_0 .

Theorem for Consistency

Suppose that $Q(\theta; X)$ is continuous in θ and there exists a function $Q_0(\theta)$ such that

1. $Q_0(\theta)$ is uniquely maximized at θ_0
2. Ω is compact
3. $Q_0(\theta)$ is continuous in θ
4. $Q(\theta; X)$ converges uniformly in probabiloty to $Q_0(\theta)$.

Then, $\widehat{\theta} = \mathit{argmax}_\theta Q(\theta; X)$ converges to $\theta_0 (= \mathit{argmax}_\theta Q(\theta))$ in probability, where $Q(\theta; X) = l(\theta; X)/n$ and $Q_0(\theta) = E_{\theta_0} Q(\theta; X)$

Proof

- For a positive ϵ, define $\epsilon-$neighborhood about θ_0:

$$\Omega(\epsilon) = \{\theta :| \theta - \theta_0 |< \epsilon\}$$

- We need to show that

$$P_{\theta_0}[\widehat{\theta} \in \Omega(\epsilon)] \to 1$$

  as $n \to \infty$.

- Since $\Omega(\epsilon)$ is an open set, $\Omega \cap \Omega(\epsilon)^C$ is a compact set (Assumption 2).
- Since $Q_0(\theta)$ is a continuous function (Assumption 3), then $sup_{\theta \in \Omega \cap \Omega(\epsilon)^C} Q_0(\theta)$ is a achieved for a θ in the compact set.

Proof

- Denote this value by θ^*, which belongs to $\Omega \cap \Omega(\epsilon)^C$.
- Since θ_0 is the unique max, let $Q_0(\theta_0) - Q_0(\theta^*) = \delta > 0$.
- For any θ, we consider two possibilities: (1) $\theta \in \Omega \cap \Omega(\epsilon)^C$, and (2) $\theta \in \Omega \cap \Omega(\epsilon)$
- For Case 1, let A_n be the event that

$$sup_{\theta \in \Omega \cap \Omega(\epsilon)^C} \mid Q(\theta; X) - Q_0(\theta) \mid < \delta/2$$

- Then, A_n implies the following result:

$$Q(\theta; X) < Q_0(\theta) + \delta/2 \leq Q_0(\theta^*) + \delta/2$$

$$= Q_0(\theta_0) - \delta + \delta/2 = Q_0(\theta_0) - \delta/2$$

Proof

- For Case $\theta \in \Omega(\epsilon)$, let B_n be the event that

$$sup_{\theta \in \Omega(\epsilon)} \mid Q(\theta; X) - Q_0(\theta) \mid < \delta/2$$

- Then, B_n implies the following result:

$$Q(\theta; X) > Q_0(\theta) - \delta/2 \text{ for all } \theta$$

Hence,

$$Q(\theta_0; X) > Q_0(\theta_0) - \delta/2$$

- By comparing the last expressions for each of the cases 1,2, we conclude that if both A_n and B_n hold then $\widehat{\theta} \in \Omega(\epsilon)$.
- But by uniform convergence, $P(A_n \cap B_n) \to 1$, so

Converges Uniformly in Probability

- This is often difficult to prove.
- (Bonus Homework) If $logp(x;\theta)$ is continuous in θ for all $\theta \in \Omega$. If there exists a function $d(x)$ such that $\mid logp(x;\theta) \mid \leq d(x)$ for θ and x, and $E_{\theta_0}[d(X)] < \infty$. Then, (1) $Q_0(\theta) = E_{\theta_0}[logp(X;\theta)]$ is continuous in θ; (2) $sup_{\theta\in\Omega} \mid Q(\theta;X) - Q_0(\theta) \mid \to 0$ in Pr.

Inconsistency of an MLE

- Let $Y_{i1}, Y_{i2} \sim N(\mu_i, \sigma^2)$, $i = 1, \ldots, n$.
- It is easy to show that the MLE of σ^2 is

$$\widehat{\sigma}^2 = \frac{1}{2n} \sum_{i=1}^{n} \sum_{j=1}^{2} (Y_{ij} - \bar{Y}_i)^2,$$

  where $\bar{Y}_i = (Y_{i1} + Y_{i2})/2$.

- We can show that

$$\widehat{\sigma}^2 \xrightarrow{P} \sigma^2/2.$$

- Hence, the MLE is not consistent. This is due the fact that the number of parameters is not fixed.

Score and Fisher Information

- The score function and Fisher information are important quantities in many aspects of statistical inference.
- Score function:

$$S_n(\theta) = l'(\theta) = \frac{\partial \log p(X_1, \ldots, X_n; \theta)}{\partial \theta} = \sum_i \frac{\partial \log p(X_i; \theta)}{\partial \theta}$$

  under the iid sampling.
- The Fisher information: $I_n(\theta) = Var_\theta(S_n(\theta))$.
- $I_n(\theta) = nI(\theta)$, where

$$I(\theta) = -E\left(\frac{\partial^2 \log p(X; \theta)}{\partial \theta^2}\right).$$

Score and Fisher Information

- For the vector case, $\theta = (\theta_1, \ldots, \theta_k)$, the Fisher information is a $k \times k$ matrix with the (r, s) element, defined by

$$I_n(\theta)(r, s) = -E\left(\frac{\partial^2 l(\theta)}{\partial\theta_r \partial\theta_s}\right).$$

Asymptotic Normality of MLE

- The regularity conditions we need to prove the asymptotic normality of the MLE are stronger than those needed for consistency.
- We need to assume that
  1. the dimension of the parameter space does not change with n,
  2. $p(x;\theta)$ is a smooth function of θ,
  3. we can interchange differentiation w.r.t. θ and integration over x, and
  4. the range of X does not depend on θ.

Asymptotic Normality of MLE

- Under the regularity assumptions,

$$\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right).$$

- That is,

$$\widehat{\theta}_n = \theta + O_P\left(\frac{1}{\sqrt{n}}\right).$$

Slutsky's Theorem

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} a$, then

- $Y_n X_n \xrightarrow{d} aX$.
- $Y_n + X_n \xrightarrow{d} a + X$.

Multivariate Delta Method

- Let $Y_1, \ldots, Y_n$ be a sequence of random vectors that satisfies that

$$\sqrt{n}(Y_n - \theta) \xrightarrow{d} N_p(0, \Sigma).$$

- Suppose $g'(\theta)$ exists and is not 0. Then,

$$\sqrt{n}(g(Y_n) - g(\theta)) \xrightarrow{d} N\left(0, \left(\frac{\partial g(\theta)}{\partial \theta}\right)' \Sigma \left(\frac{\partial g(\theta)}{\partial \theta}\right)\right).$$

Proof of Asymptotic Normality of MLE

- For Score function, we have $S_n(\widehat{\theta}) = 0$.
- Taylor expansion of $S_n(\widehat{\theta})$ about θ_0.
- Central Limit Theorem
- Weak Law of Large Numbers
- The detail is given on the blackboard.

Efficiency

Under the previously mentioned regularity conditions, the MLE is optimal. The details are very complicated so we will only discuss this heuristically.

- Under the regularity assumptions, we have

$$\widehat{\theta} = \theta + \frac{1}{n}\sum_{i=1}^{n} \psi^*(X_i) + o_P(n^{-1/2}).$$

  for the MLE.
- It can be shown that for a regular estimator (well-behaved one) $\tilde{\theta}$, we can also have

$$\tilde{\theta} = \theta + \frac{1}{n}\sum_{i=1}^{n} \psi(X_i) + o_P(n^{-1/2}).$$

Efficiency

- We can show that $Var(\psi(X)) \geq Var(\psi^*(X))$.
- Hence, we have shown that

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta)), \ \sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{d} N(0, v(\theta)),$$

where $v(\theta) \geq 1/I(\theta)$.

Relative Efficiency

If

$$\sqrt{n}(W_n - \theta) \xrightarrow{d} N(0, \sigma_W^2), \ \sqrt{n}(V_n - \theta) \xrightarrow{d} N(0, \sigma_V^2),$$

the asymptotic relative efficiency (ARE):

$$ARE(V_n, W_n) = \sigma_W^2/\sigma_V^2.$$

Robustness

- The MLE is efficient only if the model is right. The MLE can be bad if the model is wrong.
- We may also consider using nonparametric methods.
- One can also replace the MLE with estimators that are more robust.

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Example

- Suppose we assume that $X_1, \ldots, X_n \sim N(\theta, \sigma^2)$.
- MLE for θ is: $\widehat{\theta} = \bar{X}$.
- Let M_n be the sample median: $M_n = X_{(\frac{n+1}{2})}$ if n is odd and $M_n = (X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)})/2$ if n is even.
- However, if we have a perturbed model: X_i is $N(0, \sigma^2)$ with probability $1 - \delta$ and is Cauchy with probability δ. Then, $Var(\bar{X}_n) = \infty$.
- Consider the median M_n.

.

Example

- Under the normal assumption:

$$ARE(M_n, MLE) = 0.64.$$

- But, under the perturbed model the median still performs well while the MLE is terrible. In other words, we can trade efficiency for robustness.

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